

Helix OTA — Phase-1 Test-Coverage Consolidated Report

Revision: 1 **Last modified:** 2026-06-23T16:00:00Z **Scope:** consolidation of the 2026-06-23 comprehensive test-coverage + anti-bluff build-out program (Phase 0 enablers + Phase 1 per-test-type real coverage). **Authority:** the live program ledger is MASTER_TEST_COVERAGE_PLAN_20260623.md (§11.4.45 — not duplicated here; this is the audit-grade snapshot of what closed). **Honesty (§11.4.6):** every claim below cites the captured-evidence path that backs it. Measured numbers are FACT; in-flight / pending items are marked explicitly. Green numbers are real only because the “Honest findings & bluffs caught” section is real — read both together.

1. Executive summary

Phase 1 of the test-coverage program is **complete** for the autonomously-reachable test types. The Go control plane and the two Android JVM bricks are exercised against the **real production-ready system** (real HTTP + real Postgres, booted on-demand via the containers brick), with every PASS backed by captured physical evidence and every functional gate paired with a §1.1 mutation that proves the gate is not a bluff.

Delivered (all committed + 4-remote synced):

- **Phase 0 enablers:** F-CLUSTER real-system boot (`system.compose.yml` ota-server + Postgres + `/readyz`), F-ANTIBLUFF-LIB (`ab_pass_with_evidence`), F-METAGATES (all 14 propagation gates + functional gates §1.1-paired).
- **Phase 1 per-type coverage:** integration (pgx), e2e/full-automation, security (trust-boundary + fuzz + saturation/DDoS), stress/chaos, benchmark, Android JVM, server unit, ota-* Go unit, cmd/* smoke.
- **Anti-bluff layer:** the §11.4.165 independent-verifier was run on the batch and **caught + fixed a tautological restore-integrity bluff** in the meta-test harness.

Remaining (honest gaps): on-device Android A/B (needs hardware), Phase 2 (HelixQA Go orchestrator + vision against the live system; new OTA challenges), functional media/vision gates (§11.4.163 — latent for a backend with no UI surface), and the rate-limiter-ships-OFF hardening recommendation (operator decision, §11.4.122).

Commit range: d94d84f9..HEAD (HEAD = 86734699), 13 commits.

2. Per-test-type coverage table

Test type	Measured coverage / result	Evidence path
Integration (pgx/Postgres)	store 85.5% / rollout 83.1% — real podman + Postgres	docs/qa/20260623-postgres-integration/ (coverage_func_full.txt, *_pkg_coverage.log, integration_test_run.log)
e2e / full-automation (black-box vs real DB)	challenge_operational 39/39 PASS (1 SKIP); DB-persistence proof audit_logs=14, devices=2, device_groups=0 (end-of-run delete real)	docs/qa/20260623-e2e-live-system/ (challenge_operational_live_postgres_evidence.txt)
e2e signed pipeline (caller-pubkey vs live)	15/15 PASS (signed upload→release→deploy→rollout→device-poll-receives-signed-update)	docs/qa/20260623-signed-pipeline-live/SUMMARY.txt (+ step01..step15)
Security — trust boundary	4/4 PASS (accept-valid, reject-bad-sig, request-supplied-key ignored, metadata-key-field rejected)	docs/qa/20260623-trust-boundary-live/SUMMARY.txt
Security — fuzz (go test -fuzz)	4 parsers fuzzed, 0 crashers (~10.8M total execs across upload/telemetry/validate parsers; e.g. ParsePayloadProperties 2.69M execs → PASS)	docs/qa/20260623-fuzz/ (protocol_*.log, telemetry_*.validator_*.log)
Security — saturation / DDoS	cap-enabled flood → 429 shed + recover ; default-stack 360-flood → 0 5xx , legit 6/6, recover 200	docs/qa/20260623-saturation-live/ (case_a_*, case_b_SUMMARY.txt, EVIDENCE_INDEX.txt)
Stress — HTTP load (§11.4.85)	p99 14.32ms @ 5,540 req/s, zero 5xx (readyz total=166203, errors=25 non-2xx=0; p50=3.69 p90=8.54)	docs/qa/20260623-http-load-live/SUMMARY.txt (+ latency_histogram.txt)
Chaos (§11.4.85)	4/4 survive + recover : postgres-kill→reconnect (no corruption), malformed/oversized→400, 12-race idempotent register→exactly-one-device, 400-conn churn→recover	docs/qa/20260623-chaos-live/ (case_a..case_d, SUMMARY.txt)
Benchmark	benchstat baseline registry + negation-proven guard, 7 benches (e.g. MemoryFindDelta 196ns/0-allocs); thresholds calibrated on measured variance	docs/qa/20260623-benchmarks, EVIDENCE.md + docs/benchmark/ (guard run/negation/exit-codes)
Android JVM (instrumentation/ ui via JaCoCo)	ota-update-engine-bridge LINE 100% (113/113, 27 tests); ota-android-agent :core LINE 100% / BRANCH 100% (poll 77→100%, json 55.6→100%, 47 tests)	docs/qa/20260623-android-jacoco/, docs/qa/20260623-js-coverage/, docs/qa/20260623-agent-poll-coverage/ COVERAGE_EVIDENCE.md
Server unit (Go)	device 92.6% / api 90.5%	docs/qa/20260623-server-coverage/ (device_cover_func_after.txt)

Test type	Measured coverage / result	Evidence path
ota-* unit (Go)	ota-protocol 100% , ota-telemetry-schema 98.9% (and per the ledger ota-validator/rollout 100%)	api_cover_func_after.txt, coverage_evidence.md) docs/qa/20260623-ota-coverage(*_after.txt)
cmd/* + tools/* smoke	each main boots; vet clean, gofmt clean; coverage applyport 24.4%, ota-server 5.1%, ota-device-emu 0.0% , loadtest 63.3%	docs/qa/20260623-cmd-smoke/evidence.txt
Anti-bluff per-assertion helper	F-ANTIBLUFF-LIB 8/8 guard, mutation-proven (always-pass mutant caught; no-evidence-no-PASS enforced)	docs/qa/20260623-antibluff-MUTATION_PROOF.txt (+ selftest_output.txt)
Anti-bluff gates (§1.1 meta-tests)	meta suite 6/6 bluff-proof; all 14 propagation gates §1.1-paired (data-driven); semgrep fail-open FIXED; one real flake found+fixed (§11.4.50, 10/10 det)	docs/qa/20260623-metagates/, docs/qa/20260623-propagation-metagates/EVIDENCE.md, docs/20260623-indep-verif-gate/
F-CLUSTER real-system boot	boots ota-server + Postgres, /readyz→200, podman healthy	docs/qa/20260623-real-system-boot/REPORT.md (podman_ps_and_health.txt, api_smoke.txt)
Cuttlefish Tier-2 A/B (reference)	slot flip + rollback trace captured on cvd (prior proven Tier-2)	docs/qa/20260623-cuttlefish-tier2-ab/REPORT.md

3. Honest findings & bluffs caught (the anti-bluff proof)

This section is the evidence that the green numbers above are **real** — every item is a finding surfaced and recorded honestly (§11.4.6), not a paper-over.

1. **Rate-limiter ships OFF (HONEST CONFIG FINDING, not silently changed §11.4.122).** The DDoS resilience PASS is real, but the default shipped stack (server/deploy/system.compose.yml) leaves HELIX_MAX_INFLIGHT **unset**, so maxInflightMiddleware is a no-op passthrough. Case (a) proves the 429 shed/recover control **works when the cap is set**; case (b) shows the default stack survives a bounded 360-flood via the host scheduler + bounded work, **NOT** via shedding. This is recorded as a hardening recommendation, not a silent config flip. *Evidence:* docs/qa/20260623-saturation-live/case_b_SUMMARY.txt.
2. **Tautological restore-integrity bluff caught + fixed by the independent verifier (§11.4.165).** When the CM-INDEPENDENT-VERIFICATION-AGENT functional gate was added, the independent review found the meta-test harness could restore-and-pass without genuinely re-asserting integrity. The fix wired a

fatal restore-integrity abort (MT_RESTORE_FAILED flag + exit 90) so the verifier catches a bluff **in itself**. Proven byte-identical restore (sha256 BEFORE == AFTER, no residue §11.4.84); meta suite went 5/5 → **6/6**. *Evidence:* docs/qa/20260623-indep-verif-gate/README.md, restore_integrity.txt, summary_line.txt.

3. **Load-harness FAIL-bluff class.** The HTTP-load harness was hardened so a harness-internal error cannot masquerade as a product FAIL (and vice versa) — the captured run reports total/rps/errors/non_2xx as independent fields with a calibrated tail ceiling, so a PASS is anchored to real counts (errors=25 non_2xx=0), not an absence-of-error inference. *Evidence:* docs/qa/20260623-http-load-live/SUMMARY.txt.
4. **Structurally-unreachable telemetry branch.** A telemetry decode branch was identified as not reachable through the public path; rather than forcing a synthetic PASS, it is left honestly uncovered (ota-telemetry-schema **98.9%**, the 1.1% being the unreachable edge) — coverage is reported as-measured, not rounded up. *Evidence:* docs/qa/20260623-ota-coverage/ota-telemetry-schema_after.txt.
5. **ota-device-emu 0.0% — but build + exec tested.** The cmd/ota-device-emu main reports 0.0% unit coverage. This is recorded honestly: it is a dev/emulation tool, smoke-covered (it builds, boots, vet-clean, gofmt-clean) but carries no unit tests. Not claimed as covered. *Evidence:* docs/qa/20260623-cmd-smoke/evidence.txt.
6. **Real intermittent flake found + fixed (§11.4.50).** During meta-test build-out a genuine intermittent flake was discovered and root-caused (not demoted to “transient”), then proven deterministic 10/10. *Evidence:* docs/qa/20260623-metagates/.
7. **e2e self-hosting honesty.** 4/5 e2e suites are self-hosting-by-design (the test must own the artifact pubkey to sign), so a caller-pubkey F-CLUSTER mode (pipeline_signed_live.sh) was drafted to exercise the signed pipeline vs the live system — recorded as IN-FLIGHT in the ledger until its evidence dir was committed, never claimed before it landed. *Evidence:* docs/qa/20260623-signed-pipeline-live/SUMMARY.txt.

4. Remaining work (honest §11.4.6)

Item	Status	Why blocked / next step
On-device Android A/B (update_engine + AVB/dm-verity + auto-rollback)	pending	Needs real RK3588 / Orange Pi 5 Max hardware (or Cuttlefish cvd Tier-2). JVM coverage is 100%; the on-device flow is the next layer. Cuttlefish Tier-2 reference captured at docs/qa/20260623-

Item	Status	Why blocked / next step
Phase 2 — HelixQA Go orchestrator vs live system	gap	cuttlefish-tier2-ab/. cmd/helixqa orchestrator + vision toolchain (helixqa-text/recvalidate/recording-analyzer) not yet wired against the live OTA system; bank-runner gates only today.
Phase 2 — new OTA challenges (AB-G4)	partial	15 shell-dispatch banks exist; OTA-specific challenges (bad-signature-rejected, request-key-ignored, telemetry-schema, rollout-staged+halt, chaos) each need real dispatch + evidence + paired mutation. docs/qa/20260623-challenges-bank/.
Functional media/vision gates (§11.4.163)	latent for backend	helix_ota ships no UI surface yet; the §11.4.163 media-validation pipeline binds the moment a UI/recorded-artifact surface ships (§11.4.96 pattern). Recorded honestly as latent, not as a failed gate.
Rate-limiter hardening	recommendation (operator decision)	Recommend setting HELIX_MAX_INFLIGHT in the shipped system.compose.yml so the proven 429 shed/recover control is active by default. Operator-gated per §11.4.122 (capability change), not silently applied.

5. Commit range

d94d84f9...HEAD — **13 commits** (HEAD = 86734699 “test(coverage): Phase-1 batch 4 — saturation + cmd-smoke + §11.4.165 gate + agent 100%”; base d94d84f9 “docs(testing): real test-coverage audit + push-all to all upstreams”). All committed + 4-remote synced by the conductor.

This report is a faithful consolidation of the committed docs/qa/20260623- evidence corpus. It does not assert any result not backed by a cited captured-evidence path. HTML + PDF siblings (§11.4.65) are generated by the conductor, not in this doc.*